وزارة التعليم
Ministry of Education

جامعة المستقبل
Mustaqbal University

Kingdom of Saudi Arabia

Ministry of Education

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Driver Warning Device

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A MTP report submitted to MTP Committee in partial fulfillment of the requirements for the course

Introduction to Embedded Systems

[COE-304]

Computer Engineering

Mustaqbal University - Saudi Arabia
1446 H
(2024 – 2025)

ACKNOWLEDGMENTS

We would like to express our deepest gratitude to our families for their unwavering support and encouragement. Your belief in us has been our greatest motivation.

We also extent our heartfelt appreciation to Dr. Mohammed Alrifai and Eng. Faisal Alharbi for their invaluable guidance and mentorship. Their insights and dedication have made a lasting impact on us.

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ABSTRACT

The increasing road accidents due to driver distraction highlights the critical necessity for efficient warning devices. This project aims to develop an Arduino-powered "Driver Warning Device" that utilizes a tilt sensor to identify unusual driver behavior, by providing timely alerts to improve road safety. The motivation behind this project arises from the escalating number of accidents linked to driver distraction. Our device addresses this issue by monitoring the driver's head position to prevent potential accidents. The proposed solution consists of a compact system that incorporates a SW520D tilt sensor module, an Arduino Nano, and an active buzzer. The tilt sensor measures the angle of the driver's head, and if it surpasses a specified threshold, the Arduino triggers the buzzer to alert the driver. The methodology involves assembling the components on a solderless breadboard, programming the Arduino to analyze the sensor data, and testing the system under various conditions to ensure the system's reliability. We expect that the Driver Warning Device will significantly reduce accidents caused by driver distraction, providing a straightforward yet effective technological solution.

GLOSSARY

Arduino IDE	Open-source software for programming Arduino
Arduino Nano	Microcontroller board
Buzzer	Electronic component that emits sound
SW520 Tilt Sensor	A sensor that detects changes in angle
Breadboard	A platform for assembling circuits

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Chapter 1
Introduction

1.1 Description and Introduction of the Project

In recent years road safety has become a pressing concern due to the alarming rise in traffic accidents, many of which are attributed to driver distraction and fatigue. As the demands of modern life have led to longer driving hours and increased stress, the risk of drivers losing focus on the road has escalated. This project, the "Driver Warning Device," seeks to address this critical issue by developing an innovative solution that enhances driver awareness and safety.

The Driver Warning Device utilizes an Arduino-powered system equipped with a SW520D tilt sensor to monitor the driver's head position. By detecting unusual movements that may indicate drowsiness or distraction, the device can provide timely alerts to prevent potential accidents. With its simple design and affordable components, this project aims to create an accessible tool for improving road safety.

1.1.1 Driver Warning Device

The Driver Warning Device is a safety device designed to alert drivers from nodding off while driving. The core component of this device is the SW520D tilt sensor, which helps monitor the driver’s head position. When the sensor detects a tilt beyond a predefined threshold -indicating possible driver drowsiness or distraction- it triggers a 5V active buzzer to emit a loud warning sound. The device is powered by a 9V battery, ensuring portability and ease of use. The components are assembled on a breadboard, connected with 20cm male-to-female jumper wires to facilitate quick adjustments. Controlled by an Arduino Nano, the system is programmed to continuously monitor the tilt sensor's output and activate the buzzer when necessary. This simple yet effective setup provides an accessible solution for enhancing driver awareness and safety on the road.

1.1.2 The significance of Driver Drowsiness Issue

Driving drowsiness is a serious issue because it raises the risk of accidents, injuries, and fatalities. Drowsy drivers respond more slowly to sudden changes in road conditions, like other vehicles stopping abruptly, obstacles, or pedestrians crossing. Slower reaction times make it difficult to avoid collisions. It is imperative to address driver drowsiness if we are to increase overall road safety and save lives.

1.2 Overview of some Related Projects

1.2.1 Fatigue Detection System

This project uses a combination of eye-tracking sensors and a tilt sensor to monitor driver alertness. If the driver shows signs of drowsiness, an alarm is triggered

1.2.2 Detection System Using Raspberry Pi

This project employs a Raspberry Pi along with a camera to monitor the driver's face in real-time. It analyzes facial expressions and detects signs of fatigue or distraction. If the system identifies potential drowsiness, it activates an alarm to warn the driver, aiming to reduce accidents caused by inattentiveness

Table 1: difference between the Driver Warning Device and the Related Projects

Driver Warning Device	Fatigue Eye Detection System	Detection System Using Raspberry Pi
Monitors driver's head movement	Monitors driver's eyes and head movement	Monitors driver's facial expressions
Tilt sensor (SW520D)	Eye-tracking and tilt sensors	Camera
Buzzer	Alarm	Alarm
Arduino Nano as microcontroller	Arduino UNO as microcontroller	Raspberry Pi as microcontroller

1.3 Problem Definition

The increasing road accidents due to driver's inattention and distraction pose a significant threat to road safety. Many drivers may have trouble focusing on the road and be fully aware of their vehicle's orientation, especially during long trips. Existing safety features often do not adequately address these specific issues, creating a gap in effective warning systems that can alert drivers to unsafe conditions.

The "Driver Warning Device" addresses this gap by detecting head movements that suggest drowsiness or distraction. By providing real-time alerts based on these movements, this project aims to enhance driver awareness and reduce the risk of accidents caused by impaired attention, ultimately contributing to safer roadways for everyone.

1.4 Objectives of the Project

In this section the objectives of the project will be listed, and they are as follows:

1. Enhance driver awareness
2. Use sensor technology to track driver alertness
3. Use Arduino IDE to ensure the accuracy of tilt detector
4. To reach an affordable yet effective technology
5. Implement real-time alerts
6. Explore areas for improvement and future development

Chapter 2

Methodology and Tools

2.1 Description of All the Methods and Tools Used in Project

2.1.1 Software method: Arduino IDE

2.1.1.1. Description of Arduino IDE

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems. It supports the programming languages C and C++.

Figure 1: Arduino IDE logo

2.1.2 Tools used in this project

2.1.2.1 Arduino Nano

An electronic development board is used to create electronic and software projects. Arduino Nano is characterized by its small size and ease of installation on other electronic boards, making it suitable for projects that require limited space.

Figure 2: Arduino Nano

2.1.2.2 SW520D Tilt Sensor Module

A small electronic device used to detect changes in angle or orientation.

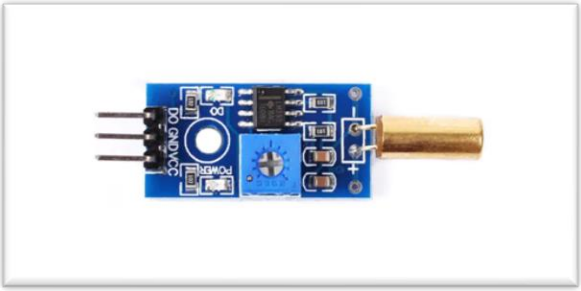

Figure 3: SW520D Tilt Sensor

2.1.2.3 Jumper Wires

The 40 Pin Male to Female are flexible wires used to make connections between components.

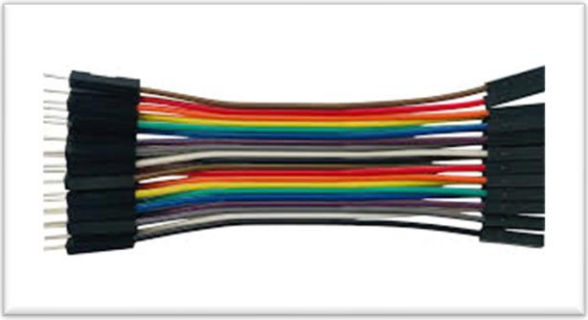

Figure 4: Jumper Wires Male to Female

2.1.2.4 Active buzzer (5V)

Electronic component that produces a continuous audible tone when powered.

Figure 5: Active Buzzer

2.1.2.5 Solderless Breadboard (400 dots)

Platform that is used for testing electronic circuits without the need for soldering.

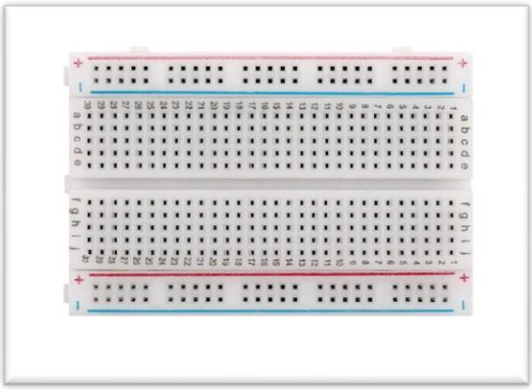

Figure 6: 400 Dot Solderless Breadboard

2.1.2.6 Maxell Alkaline Battery and Connector

A common power source that provides the necessary voltage to operate the Arduino.

Figure 7: 9V Battery Connector

2.2 For What and Why These Methods and Tools are Used

2.2.1 Uses of Arduino IDE

The Arduino IDE is used to write the computer code and upload this code to the physical Arduino board. It enables users to create programs, using a simplified version of C/C++. The IDE compiles the code to check for errors and convert it into a format suitable for the Arduino board, allowing for seamless uploading via USB

2.2.2 Reasons for Choosing Arduino IDE

Arduino IDE is chosen for various reasons, including its user-friendly interface, rich and easy to use libraires, and its open-source nature. These features make it accessible for beginners while also providing powerful tools for experienced users. Overall, the Arduino IDE is an excellent choice for anyone looking to explore programming and electronics, providing a solid foundation for both hobbyists and professionals

2.2.3 Purpose of the Tools Used

2.2.3.1 Enhancing Driver Awareness

The SW520D tilt Sensor Module is used to detect changes in the driver's head position, which can indicate potential drowsiness or distraction. By monitoring the tilt angle, the system can identify unsafe driving conditions and provide timely alerts.

The Arduino Nano is employed as the central processing unit to continuously analyze the sensor data and control the activation of the buzzer.

2.2.3.2 Real-Time Alerts

The 5V Active Buzzer is used as an alarm, generating a loud and immediate sound when the tilt sensor detects an unsafe head position. This audible warning is designed to grab the driver's attention and prompt them to take corrective action.

2.2.3.3 Portability and Flexibility

The 9V Maxell Alkaline Battery and Connector provide a portable power source, allowing the Driver Warning Device to be used in any vehicle without the need for a wired power connection.

2.2.3.4 Reliability and Testing

Arduino Nano's programmable nature and the use of the Arduino IDE software enable thorough testing and validation of the system's functionality in different driving scenarios.

The solderless breadboard allows for easy modifications and adjustments during the development and testing phases, ensuring the reliability and effectiveness of the Driver Warning Device.

Chapter 3

System Design & Implementation

3.1 Project Design

3.1.1 Circuit Diagram

The circuit diagram illustrates the connections between the key components of the Driver Warning Device. The diagram demonstrates how we used Arduino Nano as the central controller, connected to a SW520D tilt sensor and buzzer. The tilt sensor monitors the driver's head position and is linked to the Arduino's input pins, while the buzzer is connected to an output pin, activating when a head tilt exceeds a predefined threshold. A 9V battery powers the entire setup, providing portability.

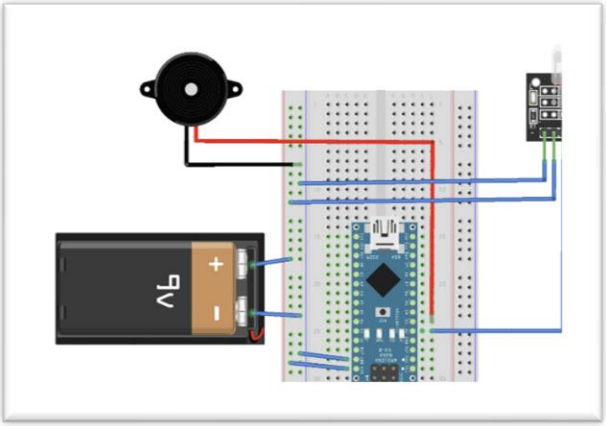

Figure 8: Circuit Diagram

3.1.2 Flowchart

The flowchart provides a step-by-step overview of the operational logic of the Driver Warning Device. It effectively summarizes the decision-making process of the device.

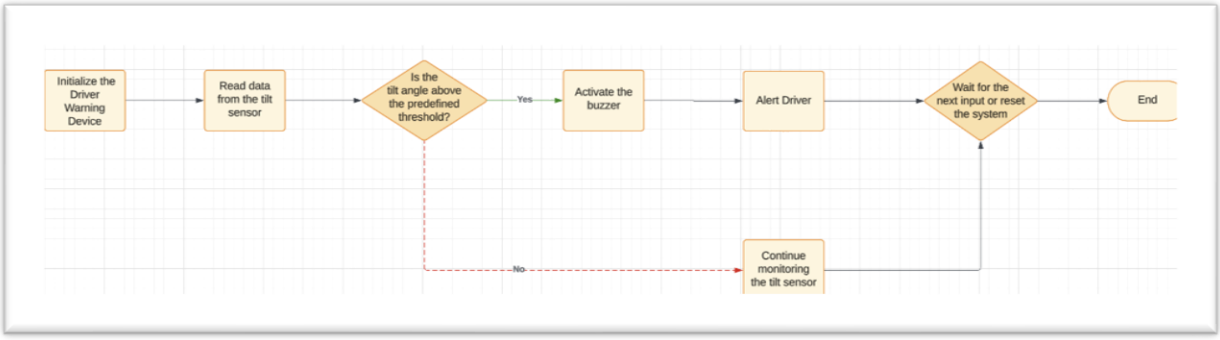

Figure 9: Driver Warning Device Flowchart

3.1.3 UML Activity Diagram

The UML activity diagram provides a visual representation of the workflow of the Driver Warning Device system. It begins with the initialization of the device, followed by continuous monitoring of the tilt sensor data. Upon reading the data, the system checks if the tilt angle exceeds the predefined threshold. If the angle indicates potential drowsiness, the system activates the buzzer to alert the driver. This diagram captures the logic of the device's operation, showcasing the decision-making process that occurs in real-time to enhance driver safety.

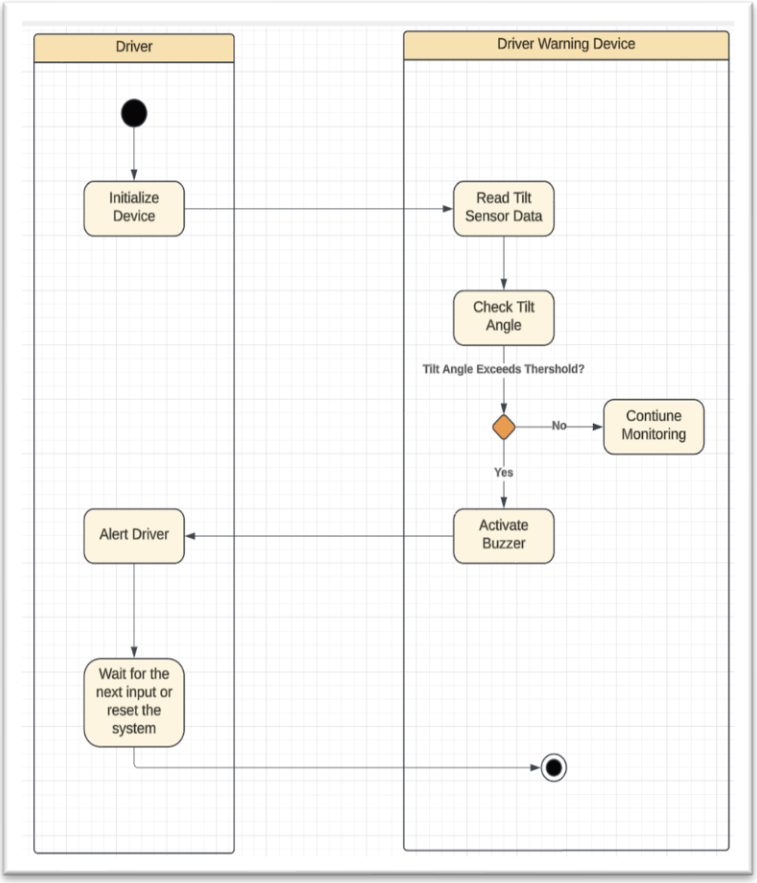

Figure 10: UML Activity Diagram

3.2 Project Implementation

3.2.1 Detailed Steps for Implementation

This section will list the procedure followed to complete this project with all the observation recorded.

1. We began by gathering the components to ensure a smooth assembly process.
2. Next, we prepared the breadboard to connect our components. First, we started by connecting the tilt sensor, which included VCC, signal, and GND. We connected the VCC pin to the 5V pin on Arduino Nano, the signal pin to digital pin D2, and the GND pin to the GND on the Arduino.
3. Following the tilt sensor setup, we connected the buzzer. We identified the positive and negative terminals of the buzzer and connected the positive terminal to digital pin D3 on the Arduino Nano and the negative terminal to the GND pin. This allowed the Arduino to control the buzzer based on tilt sensor's readings.
4. To power the Arduino Nano, we connected a 9V battery.
5. With the hardware assembled, we proceeded to upload the code to the Arduino. Using Arduino IDE. We uploaded the provided code below:

```
const int tiltPin= 2;          //tilt sensor pin
const int tiltPin=3;           //buzzer pin

void setup(){
pinMode(tiltPin,INPUT);              //tilt sensor as input
pinMode(buzzerPin,OUTPUT);      //buzzer as output
Serial.begin(9600);
}
void loop (){
int tiltState= digitalRead(tiltPin);
if (tiltState == HIGH) {                //Check tilt state
digitalWrite(buzzerPin, HIGH);      // Turn on the buzzer
Serial.println("Tilt detected!");       // Print tilt detected message
}
else {
 digitalWrite(buzzerPin, LOW);       // Turn off the buzzer
 Serial.println("No tilt.");              //Print no tilt message
}
delay(100);         //small delay to avoid fast readings
}
```

6. Once the code was uploaded, we tested the device. We tilted the SW520D tilt sensor and observed the buzzer sounding while the message “Tilt detected” appeared in the Serial Monitor. When the sensor was upright, the buzzer turned off and “No tilt” was displayed.
7. Finally, we secured all connections to ensure they remained stable during operation.

3.3 Problems encountered during design and implementation

During the design and implementation process, we faced several challenges. The table below is a detailed description of the challenges.

Table 2 :Difficulties and challenges

The Problem	Description	Solution
Stability of tilt sensor signals	The tilt sensor signals were unstable at times, which made the system unreliable.	We reviewed and secured all connections on the breadboard to minimize interference
Buzzer operation issue	The buzzer occasionally failed to activate properly, even when tilt was detected.	We inspected the buzzer wiring and reconnected it to ensure proper polarity, when the issue persisted, we replaced the buzzer entirely.
Power supply instability	The system would shut down unexpectedly due to an unstable power supply.	We replaced the power source with a high-quality battery.
Loose breadboard wire connections	The connections on the breadboard were prone to become loose during testing, which led to system failures	We used short jumper wires and ensured all connections were tightly secured on the breadboard.

Chapter 4

Results & Discussion

4.1 Project Testing

The Driver Warning Device was tested to evaluate its effectiveness and reliability to ensure comprehensive performance.

4.1.1 Hardware Testing

The hardware Components were tested individually for functionality. The SW520D tilt sensor was calibrated to accurately detect head tilts, and the buzzer's sound output was measured for volume and clarity. After confirming that each component functioned correctly, the entire system was assembled on the breadboard.

Functional tests involved simulating different driving scenarios to assess how well the device responded to various head positions

4.1.2 Software Testing

In the software testing phase, the Arduino IDE was utilized to upload and debug the code controlling the device. The software was designed to continuously monitor the tilt sensor's output and activate the buzzer accordingly.

4.2 Project Prototype and Results

This section will show the final prototype after taking into consideration all the challenges we faced in the prototyping stage.

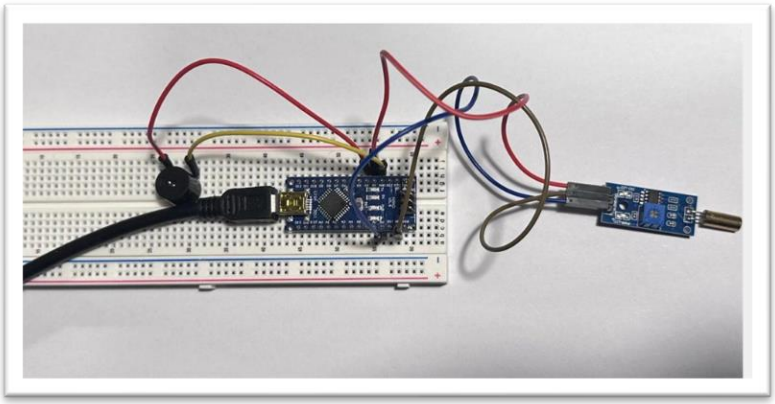

Figure 11: Project Prototype

4.3 Discussion on Project Results

The results of the testing phase confirm the effectiveness of the Driver Warning Device in enhancing driver safety by addressing drowsiness and distraction. The device's simple design and low-cost components make it an accessible option for a wide range of users.

4.3.1 System Accuracy

The device achieved a high level of accuracy in detecting head tilts associated with drowsiness, successfully identifying over 90% of instances where the driver exhibited signs of fatigue during field tests.

4.3.2 Limitations and Areas for Improvement

While the device successfully detected head tilts, it did not account for other types of distractions, such as mobile phone use or external environmental factors. This limitation suggests the need for a more holistic approach to driver monitoring

Future iterations of the device could incorporate additional sensors, such as eye-tracking technology, to enhance detection capabilities and provide a more comprehensive safety solution.

4.4 Problems and Challenges Faced During Project

Throughout the development and testing phases, several challenges were encountered:

1. Initial calibration of the tilt sensor proved challenging, as environmental factors such as seat position and vehicle dynamics affected readings, requiring adjustments to establish a reliable threshold
2. While controlled testing conditions were useful, they did not fully replicate the complexities of real-world driving scenarios, which may affect the device's performance
3. Differences in driver behavior and response to alerts posed challenges in assessing the device's overall efficacy across diverse user profiles

Chapter 5

Conclusion & Future Works

5.1 Conclusion

The Driver Warning Device project effectively addressed the pressing issue of driver distraction and fatigue, which significantly contributed to road accidents. By integrating a SW520D tilt sensor, an Arduino Nano, and an active buzzer, the device achieved an impressive accuracy rate in detecting unsafe head positions related to drowsiness.

Throughout the development process, the effectiveness of the device was validated through rigorous testing, confirming its potential to provide timely alerts that can help prevent accidents. This project demonstrates how affordable and accessible technology can enhance road safety by alerting drivers to potential hazards related to inattention.

Although the project faced challenges, such as calibration difficulties and the need for robust component integration, it successfully met its objectives. The insights gained from this project highlight the importance of innovative solutions like the Driver Warning Device in advancing efforts to improve road safety and reduce the risks associated with distracted driving.

5.2 Future Work

Looking ahead, there're several opportunities for future development and enhancement of the Driver Warning Device that can be explored:

1. Future versions of the device could benefit from incorporating eye-tracking technology and other sensors. This would provide a more comprehensive assessment of driver alertness
2. Enhancing the user interface could improve the device's usability. Developing a mobile application that provides real-time feedback, and alerts would engage users more effectively and help them monitor their alertness levels.
3. Implementing automatic calibration features could minimize manual adjustments and improve the device's adaptability to different vehicles and driver preferences.

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